FINITENESS OF THE COHOMOLOGICAL DIMENSION OF FINITE TYPE MORPHISMS AND FINITE TYPE SCHEMES OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the two applications of the induction principle, the affine base cases and the Mayer-Vietoris steps propagating the bound from the pieces to their union.

Editorial note. The label lemma-finite-cd names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source etale-cohomology.tex –label lemma-finite-cd –start 20040 –end 20071 –local-id 3566 returned RESULT=OK (L3566 free, occupancy FREE, public interval clear). Candidate taken from lane-s4/halfB-priority.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The induction principle for Noetherian schemes. Cohomology of Schemes, the lemma allowing a statement about Noetherian schemes to be proved by checking affine schemes and the Mayer-Vietoris step for X = U union V. Supplies the induction pattern used twice, reducing the assertions to the affine case and to the case X = U union V.
- Finiteness of the cohomological dimension in the affine case. Same chapter, the lemma bounding the cohomological dimension of a finite type morphism with affine target, and the proposition bounding the cohomological dimension of an affine scheme over a field of finite cohomological dimension. Supply the base cases of the two inductions.
- Relative and absolute Mayer-Vietoris sequences. Same chapter, the relative Mayer-Vietoris sequence for a morphism and the Mayer-Vietoris sequence for the cohomology of a scheme. Supply the induction steps, since a bound valid for U, V and U cap V propagates to X = U union V.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 10646–10646. Prerequisite pointer invoked at proof line(s) 20054; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 0F0X: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19941–19941. Prerequisite pointer invoked at proof line(s) 20056; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01EB: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7176–7176. Prerequisite pointer invoked at proof line(s) 20070; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 01EC: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7223–7223. Prerequisite pointer invoked at proof line(s) 20060; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0F0V: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 19763–19763. Prerequisite pointer invoked at proof line(s) 20066; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let K be a field.*

(1) *If $f : X \to Y$ is a morphism of finite type schemes over K, then $cd(f) < \infty$.*
(2) *If $cd(K) < \infty$, then $cd(X) < \infty$ for any finite type scheme X over K.*

Proof. Proof of (1). We may assume Y is affine. We will use the induction principle of Cohomology of Schemes, Lemma 08DR to prove this. If X is affine too, then the result holds by Lemma 0F0X. Thus it suffices to show that if $X = U \cup V$ and the result is true for $U \to Y$, $V \to Y$, and $U \cap V \to Y$, then it is true for f. This follows from the relative Mayer-Vietoris sequence, see Lemma 01EC.

Proof of (2). We will use the induction principle of Cohomology of Schemes, Lemma 08DR to prove this. If X is affine, then the result holds by Proposition 0F0V. Thus it suffices to show that if $X = U \cup V$ and the result is true for U, V, and $U \cap V$, then it is true for X. This follows from the Mayer-Vietoris sequence, see Lemma 01EB. □